Arista Access Node IND-ACC-65 Routing Protocol Flap Mitigation SOP

Node: IND-ACC-65 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_7639

Description

The occurrence of BGP route churn, OSPF adjacency loss, and ISIS LSP flooding on the Arista Access Node IND-ACC-65 at the Indiranagar Exchange can be attributed to the intricate interplay of routing protocols and the underlying network architecture. In this specific scenario, the Access Node is configured to operate as an OSPF Area Border Router (ABR) and an ISIS Level-2 router, with BGP configured as the primary routing protocol for peering with external autonomous systems. The OSPF adjacency loss and ISIS LSP flooding can be triggered by a variety of factors, including but not limited to, misconfiguration of the OSPF interface, incorrect ISIS metric settings, or unexpected BGP route churn events. The Arista Access Node's hardware and software architecture, featuring the Extensible Operating System (EOS) and the proprietary Arista routing protocols, further exacerbates the complexity of the issue.

Pre-requisite Checks

- Verify that the Indiranagar Exchange is accessible and that remote hands are dispatched for assistance, if necessary.

- Check the status of the redundant links and ensure that they are functioning correctly.

- Review the Access Node's configuration to identify any potential misconfigurations or inconsistencies.

Potential Root Causes

Cause	Probability
Misconfigured OSPF interface	High
Incorrect ISIS metric settings	Medium
Unexpected BGP route churn events	Low

Diagnostic Commands & Expected Outputs

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Command: show ospf interface, Expected Output: Interface is up and running, with OSPF enabled, Description: This command verifies the status of the OSPF interface and ensures that it is functioning correctly.

Command: show isis database, Expected Output: ISIS LSP database is up-to-date and consistent, Description: This command checks the ISIS LSP database for any inconsistencies or errors.

Command: show bgp summary, Expected Output: BGP sessions are established and routes are being exchanged, Description: This command verifies the status of the BGP sessions and ensures that routes are
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being exchanged correctly.
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Resolution Steps

Step 1: Step: Verify OSPF Interface Configuration, Command: show ospf interface, Description: Verify that the OSPF interface is correctly configured and functioning as expected.

Step 2: Step: Correct ISIS Metric Settings, Command: configure terminal aisis metric , Description: Correct the ISIS metric settings to ensure that they are consistent with the network architecture.

Step 3: Step: Reset BGP Sessions, Command: clear bgp *, Description: Reset the BGP sessions to ensure that the Access Node is synchronized with the external autonomous systems.

Post-Check Verification

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Command: show ospf interface, Expected Output: Interface is up and running, with OSPF enabled, Description: Verify that the OSPF interface is correctly configured and functioning as expected.

Command: show isis database, Expected Output: ISIS LSP database is up-to-date and consistent, Description: Verify that the ISIS LSP database is consistent and up-to-date.

Command: show bgp summary, Expected Output: BGP sessions are established and routes are being exchanged, Description: Verify that the BGP sessions are established and routes are being exchanged correctly.
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Roll-back Procedures

- Step: Restore Original OSPF Configuration, Command: configure terminal undo ospf interface , Description: Restore the original OSPF interface configuration to ensure that the Access Node is synchronized with the network architecture.

- Step: Restore Original ISIS Metric Settings, Command: configure terminal undo isis metric , Description: Restore the original ISIS metric settings to ensure that the Access Node is synchronized with the network architecture.

- Step: Restore Original BGP Configuration, Command: configure terminal undo clear bgp *, Description: Restore the original BGP configuration to ensure that the Access Node is synchronized with the external autonomous systems.